

Deployment & Production Support Complete Guide for Java + React Full Stack Developers

Every Topic Linked Together — Deep Explanations, Real Organization-Level Examples,
Issues & Fixes, and Real-Time Production Scenarios

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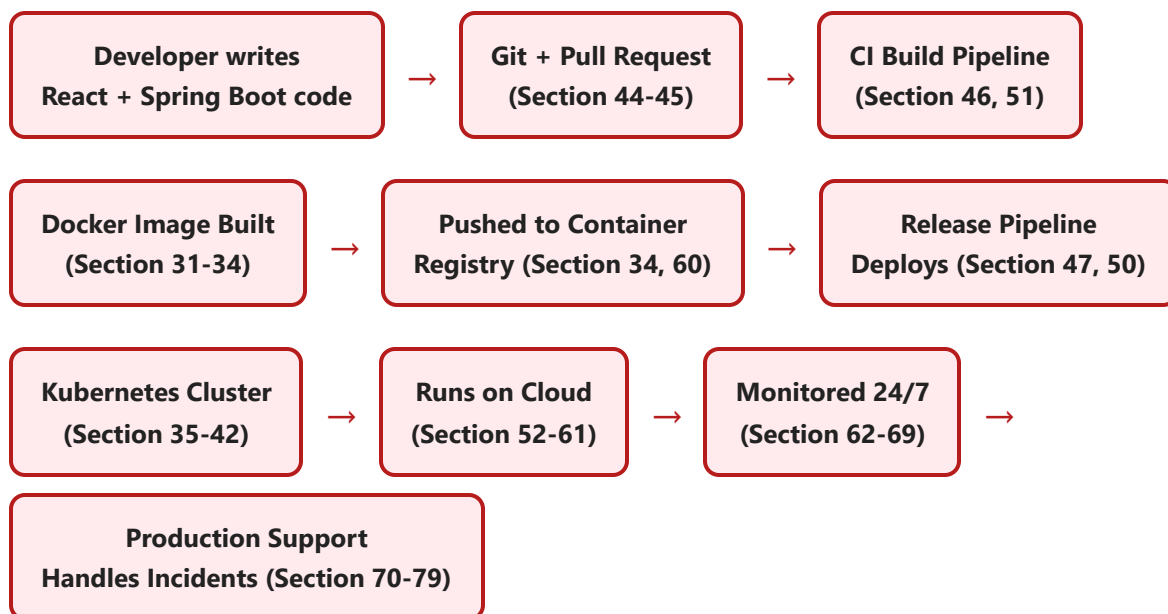
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1. Introduction — Why Every Developer Needs This Knowledge

Writing good React and Spring Boot code is only HALF of being a real full-stack developer. The other half is understanding how that code actually reaches a real user — what server it runs on, how traffic reaches it securely, how it's monitored, and what happens when something breaks at 2 AM. Developers who understand deployment and production support debug faster, design more resilient systems, and communicate far more effectively with DevOps, Cloud, and Production Support teams.

This guide links EVERY topic together — by the end, you'll see deployment not as 90 separate buzzwords, but as one connected pipeline: code → build → container → cluster → cloud → monitoring → support.

2. The Big Picture — How Every Topic in This Guide Connects



Every single topic in this guide is a real STEP in this one pipeline — this guide is organized to teach them in the order you'd actually encounter them, from your laptop to a real user's browser.

3. Recommended Learning Order

Deployment Basics → React Deployment → Java Deployment → Database Deployment → Docker → CI/CD → Kubernetes → Cloud → Logging & Monitoring → Production Support → Troubleshooting

This exact order is used as this guide's structure — each new section builds on concepts from the one before it, so reading front-to-back gives you the full connected picture rather than isolated facts.

4. Environments — DEV / QA / UAT / PROD

Environment	Purpose	Who Uses It
DEV	Active development and quick testing of in-progress code	Developers
QA	Formal functional/regression testing against a stable build	QA/Test team
UAT (User Acceptance Testing)	Business stakeholders verify the feature meets real requirements before go-live	Business users, Product Owners
PROD	The real, live environment serving real customers	Real end users; changes here are tightly controlled

Real org-level example: A banking app's new "Instant Transfer" feature is deployed to DEV daily by developers, promoted to QA once a sprint for structured testing, moved to UAT for the business/compliance team to sign off, and only THEN scheduled for a controlled PROD release with a Change Request (Section 73) and approval gate (Section 49).

ISSUE A bug is "impossible to reproduce" — it only happens in PROD, never in QA/UAT.

FIX Environments often have data/scale/configuration differences (real vs synthetic data volume, different feature flags) — investigate exactly what differs between PROD and lower environments rather than assuming the code itself is non-deterministic.

5. What Is a Server

A **server** is simply a computer (physical or virtual) that runs a program waiting to respond to requests from other computers (clients) — a Spring Boot application "running on a server" means that machine is listening on a specific port (Section 6), ready to accept HTTP requests.

Type	Description
Physical/Bare-metal server	A real, dedicated physical machine — less common today outside specialized needs
Virtual Machine (VM)	A software-emulated computer running on shared physical hardware (Section 53)
Container	A lightweight, isolated process sharing the host OS kernel (Section 32) — the modern default for deploying applications

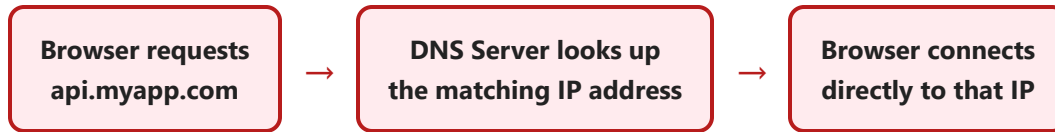
6. IP Address & Port Deep Dive

Concept	Meaning	Example
IP Address	A unique numeric address identifying a specific machine on a network	192.168.1.10 (private), 20.51.230.10 (public)
Port	A numbered "door" on that machine — a single IP can run MANY services, each listening on its own port	8080 (typical Spring Boot), 3000 (typical React dev), 5432 (PostgreSQL), 443 (HTTPS)

Real org-level example: A single VM hosting both a Spring Boot API (port 8080) and a PostgreSQL database (port 5432) — two completely separate services reachable at the SAME IP address, differentiated purely by port number.

7. DNS Deep Dive

DNS (Domain Name System) translates human-friendly domain names (`api.myapp.com`) into the actual numeric IP address a browser needs to connect to — essentially the internet's phone book.



ISSUE A newly deployed domain takes up to 24-48 hours to work for all users worldwide.

FIX DNS changes must PROPAGATE across globally distributed DNS servers, each caching results for their configured TTL (Time To Live) — this delay is expected and normal; plan domain changes well ahead of a launch date.

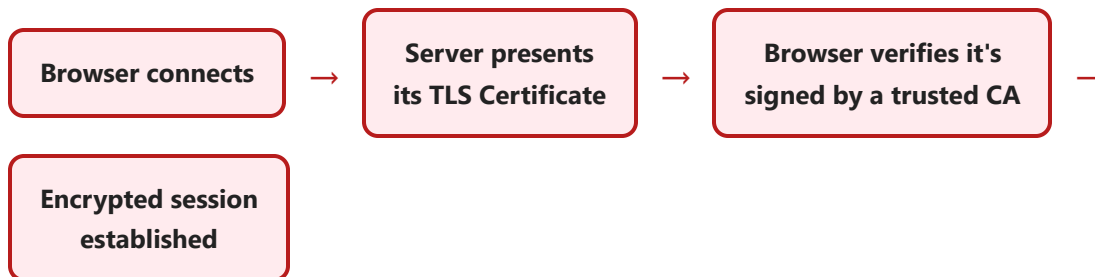
8. HTTP vs HTTPS

Protocol	Description
HTTP	Plain-text request/response protocol — anyone intercepting the network traffic can read it
HTTPS	HTTP encrypted via TLS/SSL (Section 9) — the standard, mandatory choice for any real production application today

Modern browsers actively mark plain HTTP sites as "Not Secure" and block powerful features (geolocation, camera access) on non-HTTPS pages — HTTPS is a hard production requirement, not an optional extra.

9. SSL/TLS Deep Dive

TLS (Transport Layer Security) — the modern successor to the older SSL — is the cryptographic protocol that encrypts traffic between a client and server, and verifies the server's real identity via a digital certificate issued by a trusted Certificate Authority (CA).



Real org-level example: An organization uses a managed certificate service (Azure App Service Managed Certificates, AWS Certificate Manager, or Let's Encrypt via cert-manager in Kubernetes) to automatically issue and ROTATE TLS certificates before they expire — manually renewing certificates is a classic real cause of surprise production outages when forgotten.

10. Reverse Proxy Deep Dive

A **reverse proxy** sits in front of one or more backend servers, accepting client requests on their behalf and forwarding them internally — the client only ever talks to the reverse proxy, never directly to the real backend server.

```
# Nginx as a reverse proxy in front of a Spring Boot API
location /api/ {
    proxy_pass http://localhost:8080/api;
    proxy_set_header Host $host;
}
```

Benefit	Why It Matters
SSL Termination	The proxy handles HTTPS decryption once, so individual backend services don't each need their own certificate
Hides internal architecture	Clients never see internal IPs/ports/server count — only the proxy's public address
Centralized routing	Path-based routing (e.g., /api/ → backend, / → React static files) from one entry point

11. Load Balancer Deep Dive

A **Load Balancer** distributes incoming traffic across MULTIPLE identical backend instances, so no single server handles all the load, and traffic automatically routes away from any unhealthy instance.



Real org-level example: An e-commerce company runs 6 identical Spring Boot pods behind a load balancer during Black Friday — if one pod crashes, the load balancer's health check (Section 24) detects it within seconds and stops routing traffic there, with zero visible customer impact.

12. npm in a Deployment Context

```
npm ci          # STRICT install using package-lock.json exactly – always used in CI/CD,  
never npm install  
npm run build   # produces the production-ready static files (Section 13)
```

In a CI/CD pipeline, always use `npm ci` instead of `npm install` — it guarantees the exact dependency versions from the lock file, failing fast if the lock file is out of sync, ensuring every build is byte-for-byte reproducible.

13. React Production Build Deep Dive

```
npm run build
# Produces a /dist (Vite) or /build (CRA) folder containing:
# - index.html
# - assets/index-a1b2c3.js (minified, content-hashed bundle)
# - assets/index-d4e5f6.css
```

This build process minifies code, removes dev-only warnings, tree-shakes unused code, and content-hashes filenames — the OUTPUT is a folder of plain static files with ZERO Node.js server required to run it; it just needs any static file server (Nginx, CDN).

ISSUE The app works via `npm run dev` but shows a blank page after a production build.

FIX Commonly an incorrect base path configuration, a missing environment variable only set for dev (Section 14), or an uncaught error only surfacing once dev-mode's verbose error overlay is gone — check the production build's browser Console directly.

14. Environment Variables in React

```
# .env.production
VITE_API_URL=https://api.myapp.com/api
```

Critical real-world rule: Every value placed in a React environment variable is baked directly into the PUBLIC JavaScript bundle at build time — visible to anyone via browser DevTools. Never place real secrets there; only non-sensitive config like public API base URLs belongs in frontend env variables.

15. Nginx Deep Dive

```
server {  
    listen 80;  
    root /usr/share/nginx/html;  
    location / {  
        try_files $uri /index.html;      # critical SPA fallback – Section 13's routing issue  
    }  
    location /api/ {  
        proxy_pass http://backend:8080/api/;  
    }  
}
```

Nginx is the most common real-world choice for serving a React production build — acting simultaneously as a static file server AND a reverse proxy (Section 10) forwarding API calls to the backend.

ISSUE Refreshing the browser on a client-side route (e.g., `/products/42`) returns a real 404 from Nginx.

FIX Nginx is looking for a real file at that path — add the `try_files $uri /index.html;` fallback so React Router can handle the route once `index.html` loads.

16. CORS in Production

In production, the React app and Spring Boot API are often on DIFFERENT domains/subdomains (`app.myapp.com` vs `api.myapp.com`) — CORS must be explicitly configured on the backend to allow the exact production frontend origin, just as in local development but with the REAL domain this time.

ISSUE CORS worked in every lower environment but fails specifically in production.

FIX The backend's CORS allowed-origins list almost certainly still only contains the DEV/QA frontend URLs — add the real production domain explicitly; never use a wildcard `*` origin in production if credentials/cookies are involved.

17. CDN Deep Dive

A **CDN (Content Delivery Network)** caches static files (JS, CSS, images) at edge servers physically close to users worldwide, so a user in Singapore doesn't wait on a round trip to a server in the US.

Real org-level example: A global SaaS product serves its React build through Azure Front Door/CloudFront — the first user in each region triggers an edge cache fill, and every subsequent user in that region gets the assets from the nearby edge node instead of the origin server, cutting load time dramatically.

18. Maven in Deployment

```
mvn clean package -DskipTests    # builds the deployable JAR/WAR artifact
mvn clean verify                 # full build including tests, used in CI (Section 46)
```

In a real CI/CD pipeline, Maven's `package` phase produces the exact artifact (Section 47) that gets containerized (Section 31) and deployed — the SAME artifact is promoted unchanged through DEV → QA → UAT → PROD ("build once, promote many"), never rebuilt per environment.

19. JAR vs WAR

Format	Contains	How It Runs
JAR (with Spring Boot)	Application code + an EMBEDDED Tomcat server	Standalone — <code>java -jar app.jar</code> , no external server needed
WAR	Application code only, expects a server already running	Deployed INTO an external, already-running Tomcat/application server

Modern cloud-native Spring Boot deployments (Docker/Kubernetes) almost always use the executable JAR format — WAR remains common only in legacy enterprises with existing, centrally-managed Tomcat server farms.

20. Tomcat Deep Dive

Apache Tomcat is the servlet container that actually runs a Spring Boot (or traditional Java web) application — handling incoming HTTP connections and routing them to your application code. Spring Boot bundles an EMBEDDED Tomcat by default, which is why a Spring Boot JAR can just be run directly.

Real org-level example: A legacy enterprise runs a shared, standalone Tomcat server hosting 5 different WAR applications from 5 different teams — a new Spring Boot microservice instead runs its OWN embedded Tomcat inside its own container, fully isolated from other teams' deployments (a key reason microservices moved away from shared app servers).

21. Spring Profiles

```
# application-prod.yml
spring:
  datasource:
    url: jdbc:mysql://prod-db-host:3306/proddb

java -jar app.jar --spring.profiles.active=prod
```

Spring Profiles let the EXACT SAME JAR load different configuration depending on which environment it's started in — directly implementing the "build once, promote many" principle from Section 18.

22. Environment Variables in Spring Boot

```
spring:
  datasource:
    password: ${DB_PASSWORD}    # resolved from an actual OS/container environment variable at
    startup
```

Unlike React (Section 14), backend environment variables are NEVER exposed to the public — they exist only on the server/container and are the correct place for genuine secrets (database passwords, API keys), typically injected by Kubernetes Secrets (Section 41) or a cloud secrets manager (Section 24, 59).

23. Configuration Management

Configuration Management is the discipline of externalizing every environment-specific value (URLs, feature flags, timeouts) OUTSIDE the application code, so the same build artifact behaves correctly in every environment purely through config.

Approach	Used For
application-{profile}.yml	Environment-specific Spring Boot config, bundled with the app
Kubernetes ConfigMap (Section 41)	Non-sensitive config injected at container runtime, editable without rebuilding the image
Spring Cloud Config Server	Centralized configuration served to many microservices from one Git-backed source

24. Secrets Management

Secrets (database passwords, API keys, TLS private keys) must NEVER be hardcoded or committed to source control — real organizations use a dedicated secrets manager that provides encrypted storage, access control, and audit logging.

Tool	Platform
Azure Key Vault	Azure
AWS Secrets Manager	AWS
HashiCorp Vault	Cloud-agnostic, self-hosted or managed
Kubernetes Secrets	Cluster-level, often backed by one of the above via a CSI driver

ISSUE A security scan finds a database password committed in a config file from months ago, still in Git history.

FIX Immediately ROTATE the exposed credential (changing it renders the leaked value useless) — removing it from Git history alone is not enough since it may already be cloned/cached elsewhere; then migrate to a real secrets manager going forward.

25. Health Checks

```
management:
  endpoint:
    health:
      show-details: always
# GET /actuator/health -> { "status": "UP" }
```

A **health check** endpoint lets a Load Balancer (Section 11) or Kubernetes (Section 38) automatically detect whether an instance is genuinely ready to serve traffic, pulling unhealthy instances out of rotation automatically.

26. SQL Scripts in Deployment

Raw `.sql` scripts (CREATE TABLE, ALTER TABLE, seed data INSERTs) form the foundation of database deployment — but running them manually and ad-hoc across environments is exactly the anti-pattern that Migration tools (Section 27-28) exist to prevent.

27. Database Migration Concepts

A **migration** is a versioned, ordered script describing ONE specific schema change — applied automatically, in order, exactly once, and tracked in a metadata table so every environment's schema history is fully known and reproducible.

This directly solves the real "works in QA, fails in prod because the schema is subtly different" bug class covered in the Full Stack Standards guide.

28. Flyway / Liquibase Deep Dive

```
# Flyway naming convention
V1__create_users_table.sql
V2__add_email_column_to_users.sql

mvn flyway:migrate # applies any new, not-yet-applied migrations in order
```

Tool	Style
Flyway	Plain, ordered SQL files — simple and widely used
Liquibase	XML/YAML/JSON changelogs — supports database-agnostic rollback definitions more natively

Real org-level example: On every Spring Boot startup, Flyway automatically checks its metadata table, finds any migration files not yet applied to THIS database, and runs them in order — meaning a new environment can be stood up and reach the exact correct schema version with zero manual DBA intervention.

29. Connection Pooling

Opening a brand-new database connection for every request is slow and resource-expensive — a **connection pool** (HikariCP, Spring Boot's default) keeps a set of ready-to-use connections open and reuses them across requests.

ISSUE

"Connection is not available, request timed out" errors under real production load.

FIX

Connection pool exhaustion — size the pool based on measured concurrency, and audit for any code path holding a connection too long (a slow query, a missing transaction boundary).

30. Database Rollback Strategies

Strategy	How It Works
Forward-only migrations	Never roll back a schema change — instead write a NEW migration that reverses the effect (the modern, safer standard practice)
Liquibase rollback	Explicitly defined reverse changesets, executed on demand
Backup restore	Restoring a full database snapshot taken before the change — a last resort, causing real data loss for anything written since the backup

Real production teams strongly prefer "forward-only" fixes over true rollbacks — reversing a schema change after real data has already been written against the new schema is often genuinely unsafe.

31. Dockerfile Deep Dive

```
# Multi-stage build – Section 32 explains image vs container
FROM eclipse-temurin:17-jdk AS build
WORKDIR /app
COPY . .
RUN ./mvnw clean package -DskipTests

FROM eclipse-temurin:17-jre-alpine
COPY --from=build /app/target/app.jar app.jar
EXPOSE 8080
ENTRYPOINT ["java", "-jar", "app.jar"]
```

A **Dockerfile** is a recipe describing exactly how to build a container image — the multi-stage pattern above uses a full JDK to COMPILE the app, then copies only the final JAR into a minimal runtime image, keeping the shipped image small.

32. Image vs Container

Concept	Meaning
Image	An immutable, read-only TEMPLATE built from a Dockerfile — like a class in OOP terms
Container	A RUNNING (or stopped) instance of an image — like an object instantiated from that class

```
docker build -t myapp:1.0 .      # builds an IMAGE
docker run -p 8080:8080 myapp:1.0 # starts a CONTAINER from that image
```

33. Docker Compose

```
# docker-compose.yml – runs the whole local stack with one command
services:
  backend:
    build: ./backend
    ports: ["8080:8080"]
    depends_on: [db]
  db:
    image: postgres:16
    environment:
      POSTGRES_PASSWORD: secret
  frontend:
    build: ./frontend
    ports: ["3000:80"]
```

```
docker-compose up    # starts ALL services together, on a shared internal network
```

Real org-level example: A new developer joining the team runs a single `docker-compose up` command and has the full React + Spring Boot + PostgreSQL stack running locally in minutes — no manual installation of Java, Node, or PostgreSQL needed on their own machine.

34. Container Registry

A **Container Registry** stores and versions built Docker images, ready to be pulled and run by any server/cluster — the deployment equivalent of Maven Central or npm's registry, but for entire application images instead of libraries.

```
docker tag myapp:1.0 myregistry.azurecr.io/myapp:1.0
docker push myregistry.azurecr.io/myapp:1.0
```

Registry	Platform
Azure Container Registry (ACR)	Azure
Amazon ECR	AWS
Docker Hub	Public/general-purpose

Real organizations use a **PRIVATE** registry, never a public one, for proprietary application images — combined with vulnerability scanning on every pushed image before it's allowed to be deployed.

35. Ports, Volumes & Networks in Docker

Concept	Purpose
Ports (<code>-p 8080:8080</code>)	Maps a port on the HOST machine to a port INSIDE the container, making the containerized service reachable from outside
Volumes	Persistent storage OUTSIDE the container's own filesystem — critical for databases, since a container's own filesystem is wiped when it's removed/recreated
Networks	A virtual network letting containers (e.g., backend and database) find and talk to each other by SERVICE NAME, isolated from unrelated containers

ISSUE A database container is restarted (e.g., during an image update) and ALL its data disappears.

FIX The database was writing to the container's own ephemeral filesystem instead of a mounted VOLUME — always mount a persistent volume for any stateful service's actual data directory.

36. Kubernetes Cluster & Node

Concept	Meaning
Cluster	The entire group of machines (nodes) working together, managed as one unit by Kubernetes
Node	One physical or virtual machine within the cluster, actually running containers

Docker alone runs containers on ONE machine. **Kubernetes** orchestrates containers across MANY machines (nodes) — handling scheduling, scaling, self-healing, and networking across the entire cluster automatically.

37. Pod Deep Dive

A **Pod** is the smallest deployable unit in Kubernetes — usually wrapping ONE container (occasionally a few tightly-coupled ones), with its own IP address inside the cluster.

```
apiVersion: v1
kind: Pod
metadata: { name: order-service }
spec:
  containers:
    - name: order-service
      image: myregistry.azurecr.io/order-service:1.0
      ports: [{ containerPort: 8080 }]
```

In real practice, you rarely create bare Pods directly — you define a Deployment (Section 38), which manages Pods for you, including automatically recreating one if it crashes.

38. Kubernetes Deployment Object

```
apiVersion: apps/v1
kind: Deployment
metadata: { name: order-service }
spec:
  replicas: 3 # Section 39 – how many identical Pods to keep running
  selector: { matchLabels: { app: order-service } }
  template:
    metadata: { labels: { app: order-service } }
    spec:
      containers:
        - name: order-service
          image: myregistry.azurecr.io/order-service:1.0
```

A Kubernetes **Deployment** declares the DESIRED state (how many replicas, which image) — Kubernetes continuously works to keep the ACTUAL state matching it, automatically replacing crashed Pods and managing rollouts (Section 42).

39. Kubernetes Service Object

Pods are constantly created/destroyed/rescheduled, each getting a NEW IP address every time — a **Service** provides one STABLE internal name/address that automatically load-balances across whichever Pods currently match its label selector, regardless of how many exist or how often they change.

```
apiVersion: v1
kind: Service
metadata: { name: order-service }
spec:
  selector: { app: order-service }
  ports: [{ port: 80, targetPort: 8080 }]
```

40. Ingress Deep Dive

An **Ingress** manages EXTERNAL HTTP(S) access into the cluster, routing by hostname/path to the correct internal Service — acting as the cluster's reverse proxy/API Gateway layer, letting many services share one external entry point instead of each needing its own Load Balancer.

```
apiVersion: networking.k8s.io/v1
kind: Ingress
spec:
  rules:
    - host: api.myapp.com
      http:
        paths:
          - path: /orders
            backend: { service: { name: order-service, port: { number: 80 } } }
```

41. Replicas & Scaling

```
kubectl scale deployment order-service --replicas=10    # manual scaling

# Horizontal Pod Autoscaler – automatic scaling based on real CPU usage
apiVersion: autoscaling/v2
kind: HorizontalPodAutoscaler
spec:
  minReplicas: 3
  maxReplicas: 20
  metrics: [{ type: Resource, resource: { name: cpu, target: { averageUtilization: 70 } } }]
```

Real org-level example: During a flash sale, the Order Service's HPA automatically scales from 3 to 18 Pods as CPU utilization crosses 70%, then scales back down to 3 once traffic normalizes a few hours later — with zero manual intervention from the on-call engineer.

42. ConfigMap & Secret

Object	Purpose
ConfigMap	Non-sensitive configuration (feature flags, external URLs) injected into Pods at runtime, editable without rebuilding the image
Secret	Sensitive values (passwords, tokens) — base64-encoded by default (NOT truly encrypted at rest unless etcd encryption or an external secrets manager is also configured, Section 24)

43. Rolling Update & Rollback in Kubernetes

```
kubectl set image deployment/order-service order-service=myregistry.azurecr.io/order-service:2.0
kubectl rollout status deployment/order-service
kubectl rollout undo deployment/order-service      # instant rollback to the previous version
```

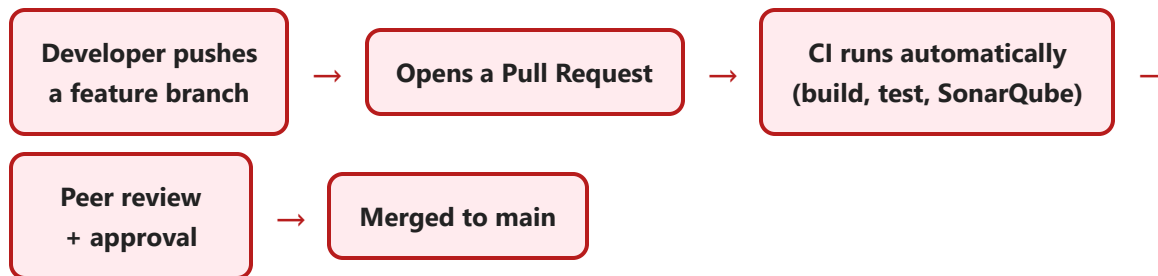
A **Rolling Update** replaces old Pods with new ones gradually, keeping the service available throughout — new Pods only receive traffic once they pass their readiness check, and old Pods are only removed once enough new ones are healthy.

Real org-level example: A bad deployment causes new Pods to fail their readiness probe — Kubernetes automatically PAUSES the rollout, keeping the old, still-healthy Pods serving all traffic; the on-call engineer runs `kubectl rollout undo` for an instant, one-command rollback to the last known-good version.

44. Git in a CI/CD Context

Every CI/CD pipeline (Section 46) is TRIGGERED by a Git event — a push to a branch, or a merged Pull Request — making Git the literal starting point of the entire deployment pipeline described in Section 2's big picture.

45. Pull Request Workflow



A Pull Request is both a CODE REVIEW gate and an automated QUALITY gate (CI checks must pass) before code is allowed to merge — branch protection rules enforce that neither can be skipped.

46. Build Pipeline Deep Dive

```
# Simplified Azure Pipelines build stage
steps:
- script: mvn clean verify          # compile + run tests
- script: npm ci && npm run build    # frontend build
- task: Docker@2                    # build and push the container image
```

A **Build Pipeline** automatically compiles, tests, and packages code into a deployable artifact EVERY time code changes — the automated foundation that makes "build once, promote many" (Section 18) actually possible.

47. Release Pipeline Deep Dive

A **Release Pipeline** takes the artifact the Build Pipeline produced and deploys it through each environment in sequence (DEV → QA → UAT → PROD), often with manual approval gates (Section 49) before sensitive environments.

Build and Release are deliberately SEPARATE pipelines in mature organizations — the build happens once; the exact same tested artifact is then released/promoted, never rebuilt per environment.

48. Jenkins vs Azure DevOps vs GitHub Actions

Tool	Model	Best Fit
Jenkins	Self-hosted, you manage the server and huge plugin ecosystem	Teams wanting maximum control/customization, willing to maintain infrastructure
Azure DevOps Pipelines	Managed service, deeply integrated with Azure	Organizations standardized on Azure and Microsoft tooling
GitHub Actions	Managed service, built directly into GitHub	Teams already using GitHub, wanting zero extra infrastructure

49. SonarQube Deep Dive

SonarQube automatically scans code for bugs, security vulnerabilities, code smells, and test coverage — enforced as a **Quality Gate** in the build pipeline that can BLOCK a merge/deployment if code quality drops below an agreed threshold.

Real org-level example: A team's Quality Gate requires 80% test coverage on new code and zero new critical vulnerabilities — a Pull Request introducing a SQL injection risk is automatically blocked from merging, caught before a human reviewer even looks at it.

50. Artifact Management

An **artifact** is any build OUTPUT passed between pipeline stages — a JAR, a Docker image, a test report. Artifact repositories (Azure Artifacts, Nexus, JFrog Artifactory) version and store these outputs, ensuring the EXACT SAME artifact tested in QA is the one deployed to PROD.

51. Deployment Approval Gates

Environment	Typical Approver
QA	Usually automatic
UAT	QA Lead / Product Owner
Production	Release Manager, sometimes requiring a 2-person sign-off plus a Change Request (Section 73)

Approval gates are deliberately configured in the PIPELINE PLATFORM's settings, not in pipeline code itself, so a developer changing pipeline YAML cannot silently remove a required approval step.

52. CI/CD Rollback

```
# Re-deploying a previous, known-good artifact version is usually FASTER and safer than a Git  
revert + rebuild  
kubectl rollout undo deployment/order-service
```

The fastest, safest rollback in a mature pipeline is re-deploying the PREVIOUS artifact that's still sitting in the artifact repository/container registry — not reverting code and waiting for an entirely new build to complete under pressure during an incident.

53. Azure vs AWS Basics

Concept	Azure	AWS
Virtual Machines	Azure VMs	EC2
App hosting (PaaS)	App Service	Elastic Beanstalk
Managed Kubernetes	AKS	EKS
Object Storage	Blob Storage	S3
Secrets	Key Vault	Secrets Manager

Both platforms solve the exact same core problems — the concepts transfer directly; only the product names differ, as covered in full depth in the AWS/Azure Cloud Architecture guide.

54. Virtual Machines

A cloud **Virtual Machine** is a full, software-emulated computer you configure and manage yourself (OS patches, installed software) — the closest cloud equivalent to owning a physical server, offering maximum control but the most operational responsibility.

Real org-level example: A legacy batch-processing job requiring a specific, older OS version and custom-installed software that won't run in a container is hosted on a VM, while newer microservices around it run on AKS/Kubernetes — a common real-world hybrid, not an all-or-nothing choice.

55. App Services (PaaS)

Platform-as-a-Service offerings (Azure App Service) let you deploy a JAR/WAR or container directly without managing the underlying OS/VM at all — the cloud provider handles patching, scaling infrastructure, and load balancing for you.

Responsibility	VM (IaaS)	App Service (PaaS)
OS patching	You	Cloud provider
Scaling infrastructure	You configure manually	Built-in, configuration-only
Deployment unit	Whatever you install	A JAR/WAR or container image

Real org-level example: A mid-size company migrated a Spring Boot monolith from a self-managed VM to Azure App Service specifically to eliminate the recurring monthly OS-patching maintenance window that previously required an engineer on standby every time.

56. Cloud Storage

Object storage (Blob Storage/S3) holds unstructured files (images, backups, logs) accessed over HTTP(S) — commonly paired with a CDN (Section 17) for serving static frontend assets globally.

Real org-level example: An e-commerce platform stores product images in Blob Storage, serves them through a CDN edge cache, and separately stores nightly database backups in a different, access-restricted storage container with a defined retention/lifecycle policy that automatically archives older backups to cheaper cold storage.

57. Cloud Database Services

Managed database services (Azure SQL, AWS RDS) handle backups, patching, and failover automatically — removing the operational burden of manually administering a database server, though a DBA team (Section 81) still owns schema design and performance tuning.

Real org-level example: A managed database service automatically fails over to a standby replica within seconds when the primary instance's underlying hardware experiences an issue — the application only sees a brief connection blip, rather than the extended manual recovery a self-hosted database might require.

58. Cloud Networking

Virtual Networks (VNet/VPC), subnets, and Network Security Groups define which resources can talk to which — a core security principle is keeping databases in PRIVATE subnets, never directly reachable from the public internet.

ISSUE A security audit finds a production database with a public IP address directly reachable from the internet.

FIX Move the database into a private subnet reachable only from application servers within the same trusted network, removing its public IP entirely — authentication alone should never be a database's only line of defense.

59. Container Registry (Cloud)

Covered in Section 34 — cloud-native registries (ACR/ECR) integrate directly with the cloud's IAM (Section 61) for secure, keyless authentication from CI/CD pipelines and Kubernetes clusters running in the same cloud account.

60. Managed Kubernetes Service (AKS/EKS)

Running Kubernetes yourself means managing the CONTROL PLANE (the brain coordinating the cluster) — a huge operational burden. Managed services (AKS/EKS) handle the control plane for you, letting your team focus purely on the application workloads (Pods, Deployments, Services from Sections 36-43).

Real org-level example: A team previously running self-managed Kubernetes spent significant engineering time each quarter upgrading the control plane version and troubleshooting etcd issues; migrating to AKS eliminated that entire category of work, letting the same engineers focus purely on application-level reliability instead.

61. Key Vault & Secrets Management (Cloud)

Covered in depth in Section 24 — cloud-native Key Vault/Secrets Manager services integrate with IAM (below) so a running application's Managed Identity can retrieve a secret with zero hardcoded credentials anywhere in its configuration.

62. IAM Deep Dive

Identity and Access Management controls exactly WHO (human or application) can do WHAT to which cloud resource — enforcing the principle of least privilege (grant only the minimum permission actually needed).

ISSUE A security audit finds a developer's personal account has full Owner/Administrator access to the production cloud subscription.

FIX Apply role-based access control (RBAC) granting only the specific, minimum roles each person/team genuinely needs — production infrastructure changes should flow through reviewed pipelines (Section 47), not direct manual console access.

63. Application Logs

```
log.info("Order {} created for customer {}", traceId={}, orderId, customerId,  
MDC.get("traceId"));
```

Structured application logs (Section 64) are the PRIMARY tool for understanding what an application actually did, especially after an incident — every log line should include a correlation/trace ID (covered in the Observability guide) to reconstruct one request's full journey.

64. Splunk / ELK Stack

Both centralize logs from every service instance into one searchable platform — critical since a real production system runs many replicas (Section 41), making SSH-ing into individual servers to read local log files completely impractical.

Tool	Style
Splunk	Powerful commercial platform with its own query language (SPL)
ELK (Elasticsearch, Logstash, Kibana)	Popular open-source alternative

65. Azure Monitor

Azure's umbrella monitoring platform aggregating metrics, logs, and alerts (Section 68) from every Azure resource (App Services, AKS, databases) into unified dashboards — the Azure-native equivalent covering the same purpose as Prometheus/Grafana (Section 67) in a cloud-agnostic setup.

66. Application Insights

Microsoft's Application Performance Monitoring (APM) tool — auto-instruments a Spring Boot/React app to capture distributed traces, dependency call durations, and exceptions, letting you see a complete request's timeline across frontend → API Gateway → backend → database in ONE view.

67. Grafana

The standard open-source dashboarding tool, typically visualizing metrics collected by Prometheus — teams build dashboards tracking Golden Signals (latency, traffic, errors, saturation) per service, watched continuously by on-call engineers and Production Support (Section 79).

68. Alerts Deep Dive

An **alert** automatically notifies a team (via PagerDuty, Teams, email) when a metric crosses a defined threshold — the difference between discovering an outage from a dashboard versus from a customer complaint.

ISSUE The on-call team suffers "alert fatigue" and starts ignoring pages.

FIX Alert only on genuinely user-impacting SYMPTOMS (error rate, latency breaching SLA, Section 74) rather than every internal metric blip — attach a clear runbook to every alert so responders know exactly what to check first.

69. CPU/Memory Monitoring

Tracking CPU and memory usage per instance/Pod reveals resource exhaustion BEFORE it causes an outage — directly feeding into autoscaling decisions (Section 41) and capacity planning.

70. API Monitoring

Tracking per-endpoint latency, error rate, and throughput reveals exactly WHICH API is degrading, often before broader infrastructure metrics show any problem at all — the most direct signal of real user-facing impact.

71. ServiceNow Deep Dive

ServiceNow is the most widely used enterprise platform for tracking Incidents (Section 72), Problems (Section 76), and Change Requests (Section 77) — the system of record production support and operations teams live in daily.

Real org-level example: An alert fires automatically creating a ServiceNow Incident ticket, assigned to the on-call engineer; every troubleshooting action and eventual root cause is logged directly in that ticket, forming the audit trail referenced later during the RCA (Section 78).

72. Incident Management



Incident Management is the structured process of restoring service as FAST as possible — the priority during an active incident is mitigation (stop the bleeding, often via rollback), with deep root-cause investigation deliberately deferred to the follow-up RCA process.

73. Priority Levels — P1/P2/P3/P4

Priority	Meaning	Typical Response Time
P1 (Critical)	Complete outage or severe business impact, affecting all/most users	Immediate, 24/7, all-hands response
P2 (High)	Major feature broken, significant but partial impact	Urgent, within hours
P3 (Medium)	Minor feature issue, workaround available	Next business day
P4 (Low)	Cosmetic issue, minimal impact	Scheduled into normal backlog

Real org-level example: A payment gateway completely down during business hours is a P1, triggering an immediate war-room bridge call with Devs, DBA, and Cloud teams; a misaligned button on a rarely-used admin page is a P4, simply logged as a backlog ticket.

74. SLA Deep Dive

A **Service Level Agreement** is a formally agreed commitment on system availability/performance — e.g., "99.9% uptime" (allowing roughly 8.7 hours of downtime per year) — directly driving architecture decisions like Disaster Recovery strategy and alert thresholds (Section 68).

75. Problem Management

While Incident Management focuses on fast mitigation, **Problem Management** investigates the UNDERLYING root cause of a recurring incident pattern, aiming to prevent it from ever happening again — often producing a permanent fix tracked as its own separate piece of work.

76. Change Request Process

A **Change Request (CR)** is the formal, reviewed proposal to make a change to a production system — required for any significant production deployment, requiring documented rollback plans and stakeholder approval BEFORE the change window.

77. Root Cause Analysis (RCA)

After a significant incident is resolved, an **RCA** document formally answers: what happened, why it happened (the true underlying cause, not just the immediate trigger), and what concrete actions will prevent recurrence.

Real org-level example: An RCA for a checkout outage traces the IMMEDIATE trigger (a slow third-party payment API) back to the TRUE root cause (no circuit breaker/timeout was configured on that specific outbound call) — the action item isn't "contact the payment provider" but "add a circuit breaker to every outbound integration," preventing the same CLASS of incident from a different dependency in the future.

78. Hotfix Process

A **hotfix** is an urgent, expedited fix deployed OUTSIDE the normal release cadence to resolve a live P1/P2 incident — typically following a lightweight but still reviewed pipeline (skipping non-essential slow stages like full E2E suites, while keeping critical checks like unit tests and security scans).

79. Production Troubleshooting Methodology

1. Confirm and scope the impact (which users, which feature, since when).
2. Check recent changes first (a recent deployment is the most common real cause of a NEW production issue).
3. Check dashboards/APM (Sections 65-70) for the affected service's Golden Signals.
4. Check logs (Section 63-64) filtered by the relevant time window and correlation ID.
5. Mitigate first (rollback, Section 43/52) — investigate the deep root cause AFTER service is restored.

80. Rollback in a Production Support Context

During an active incident, rollback is almost always the FASTEST safe mitigation — re-deploying the last known-good artifact (Section 43/52) takes minutes, while diagnosing and forward-fixing a complex bug under pressure can take hours the business may not have.

81. Who Maintains What — Team Roles & Responsibilities

Team	Owns
Developers	Application code, unit/integration tests, fixing application-level bugs
DevOps Team	CI/CD pipelines, build/release automation, deployment tooling
Cloud Team	Cloud account/subscription structure, networking, cost governance
Production Support Team	First response to incidents, initial triage, escalation to the right team
SRE Team	Reliability engineering — SLOs, capacity planning, automating away repetitive operational toil
DBA Team	Database schema design review, performance tuning, backup/recovery strategy
Network Team	VPNs, firewalls, DNS, load balancer/network-level configuration
Security Team	Vulnerability scanning, IAM policy, security incident response, compliance
QA Team	Test planning, functional/regression/performance testing before release
Release Management Team	Coordinating release schedules, Change Request approvals, go/no-go decisions

In smaller organizations, one person often wears several of these hats; in large enterprises, each is a genuinely separate team — understanding this map helps you know exactly WHO to escalate to (and how to communicate with them) when something goes wrong outside your own direct control.

82. Real Organization Scenario 1 — End-to-End Deployment of a New Feature

Scenario: A team ships a new "Order Tracking" feature end to end.

1. Developer writes the React component and Spring Boot endpoint, opens a Pull Request (Section 45).
2. CI Build Pipeline (Section 46) runs Maven tests, npm build, and a SonarQube quality gate (Section 49).
3. PR approved and merged; a new Docker image is built and pushed to the Container Registry (Section 34).
4. Release Pipeline deploys the new image to DEV automatically, then QA after tests pass.
5. Product Owner signs off in UAT; a Change Request (Section 76) is raised for the PROD deployment window.
6. On approval, Kubernetes performs a Rolling Update (Section 43) in production — zero downtime.
7. Application Insights/Grafana dashboards are watched closely for the first hour post-release for any anomaly.

83. Real Organization Scenario 2 — Production Incident from Alert to RCA

1. Grafana alert fires: API error rate crosses 5% — a ServiceNow P1 Incident (Sections 68, 71, 73) is auto-created.
2. On-call engineer checks recent deployments first (Section 79) — a release went out 20 minutes earlier.
3. Logs (Splunk) filtered by correlation ID reveal a null-pointer exception in the new code path.
4. Decision made to ROLL BACK immediately (Section 80) rather than attempt a forward fix under pressure — service restored within 8 minutes.
5. Incident is closed; a full RCA (Section 77) is scheduled within 48 hours, identifying a missing null check and a gap in test coverage as the true root causes, with both added as concrete action items.

84. Real Organization Scenario 3 — A Failed Release & Rollback

A database migration (Section 27-28) bundled with a release locks a large table for several minutes in production (unlike the much smaller staging table), causing widespread request timeouts. The team's runbook specifies FORWARD-ONLY migrations (Section 30) — rather than attempting a risky schema rollback, they mitigate by scaling up read replicas temporarily and communicate an ETA to stakeholders, while a follow-up migration using an online, batched schema-change approach is prepared and deployed during the next low-traffic window.

85. Real Organization Scenario 4 — Scaling for a Traffic Spike

Ahead of a planned marketing campaign, the SRE team (Section 81) load-tests the checkout service at 5x expected peak traffic, tunes the Horizontal Pod Autoscaler's thresholds (Section 41) based on real observed behavior, and verifies the database connection pool (Section 29) is sized appropriately for the increased concurrent load. During the actual campaign, Pods scale from 4 to 22 automatically as traffic rises, and scale back down afterward — with Grafana dashboards (Section 67) confirmed healthy throughout by the on-call team.

86. Common Issues & Fixes Across Deployment & Production (Extended)

ISSUE A deployment succeeds according to the pipeline, but the application is actually down.

FIX The pipeline only confirmed the DEPLOY step completed, not that the app is genuinely healthy — always gate traffic cutover on a real health check (Section 25) passing, not just deployment completion.

ISSUE "It works in my local Docker container but fails in the cluster."

FIX Commonly a missing ConfigMap/Secret (Section 42) in the cluster environment, or a Kubernetes NetworkPolicy blocking traffic that a local Docker network allowed freely — compare configuration between the two environments directly.

ISSUE A certificate expires unexpectedly, causing a sudden HTTPS outage.

FIX Use auto-renewing managed certificates (Section 9) and set up an alert (Section 68) for certificates approaching expiry well in advance — manual certificate renewal is a classic, entirely preventable outage cause.

ISSUE A team can't tell which of 15 Pods is experiencing a problem during an incident.

FIX Ensure every log line includes the Pod name/instance ID and a correlation ID (Section 63), and use distributed tracing (Section 66) to pinpoint exactly which instance/hop is genuinely misbehaving.

87. Interview & Scenario Questions & Answers (30+)

Q1. Walk through what happens from a developer merging code to it running in production.

A: Merge triggers a Build Pipeline (compile, test, containerize, quality gate) producing a versioned artifact; a Release Pipeline then promotes that SAME artifact through DEV→QA→UAT→PROD, gated by approvals, ultimately performing a Kubernetes rolling update with health checks confirming success at every step.

Q2. Why is "build once, promote many" preferred over rebuilding per environment?

A: Rebuilding per environment risks subtly different artifacts reaching production than what was actually tested in QA/UAT — promoting the exact same tested binary guarantees what you validate is genuinely what ships.

Q3. What's the difference between a Kubernetes Deployment and a Service?

A: A Deployment manages the actual Pods (desired replica count, rollout strategy); a Service provides one stable network address that load-balances across whichever Pods currently exist, regardless of how often they're replaced.

Q4. Why should database schema changes go through Flyway/Liquibase instead of manual SQL?

A: Versioned migrations guarantee every environment applies the exact same changes in the exact same order, are reviewable in source control, and avoid the classic "works in QA, fails in prod" schema-drift bug.

Q5. Why is rollback usually preferred over a forward fix during an active P1 incident?

A: Rollback to a known-good previous artifact is fast and low-risk; diagnosing and safely forward-fixing a complex bug under time pressure risks introducing a second, possibly worse issue while the business is already impacted.

Q6. What's the real difference between Incident Management and Problem Management?

A: Incident Management prioritizes restoring service FAST (mitigation first); Problem Management investigates the true underlying root cause afterward to prevent recurrence — different goals, intentionally handled at different times.

Q7. Why can't CORS ever be fixed from the frontend?

A: CORS is enforced by the browser checking the SERVER's response headers — only the backend can grant permission for a given origin; no frontend code change can bypass this browser-side security check.

Q8. How would you diagnose "connection pool exhausted" errors in production?

A: Check current pool utilization metrics, identify whether load genuinely exceeds the configured pool size or whether connections are being held too long (a slow query or a missing transaction boundary leaking a connection), then size the pool or fix the leak accordingly.

Q9. Why do organizations use a private Container Registry instead of a public one?

A: To keep proprietary application images confidential, scan them for vulnerabilities before deployment, and

integrate authentication with the cloud's own IAM system rather than trusting a public, shared registry.

Q10. What's the purpose of a Horizontal Pod Autoscaler?

A: It automatically increases/decreases the number of running Pod replicas based on real-time metrics like CPU utilization, letting a service handle a traffic spike without manual intervention and scale back down afterward to save cost.

Q11. Why should secrets never be stored as plain Kubernetes Secrets alone in a high-security environment?

A: Kubernetes Secrets are only base64-ENCODED by default, not truly encrypted at rest unless etcd encryption is also enabled — real production setups typically integrate an external secrets manager (Key Vault) via a CSI driver for genuine encryption and rotation.

Q12. What is the purpose of a readiness/health check in a load-balanced deployment?

A: It lets the load balancer or Kubernetes automatically detect and stop routing traffic to an unhealthy/not-yet-ready instance, preventing users from hitting a broken or still-starting-up Pod.

Q13. Why is a Change Request required for production deployments in many organizations?

A: It formalizes review and approval BEFORE a risky change happens, ensures a documented rollback plan exists, and creates an audit trail — reducing the chance of an unreviewed, undocumented change causing an outage.

Q14. How would you explain the difference between a P1 and a P3 incident to a new team member?

A: P1 is a critical, widespread outage demanding an immediate, full response; P3 is a minor issue with an available workaround that can wait for the next business day — priority reflects actual business impact and urgency, not just technical severity.

Q15. Why does a React SPA need special web server configuration (like Nginx's `try_files`) to work correctly in production?

A: Client-side routes only exist within the browser's React Router, not as real files on the server — without a fallback to `index.html` for unmatched paths, refreshing on a deep link returns a genuine 404 from the web server.

Q16. What's the value of a centralized logging platform like Splunk/ELK over reading individual server log files?

A: A real production system runs many replicas across many nodes — centralizing logs makes it possible to search across ALL instances by correlation ID in one place, instead of impractically SSHing into each server individually.

Q17. Why would a team choose a managed Kubernetes service (AKS/EKS) over self-managing Kubernetes?

A: Managing the Kubernetes control plane yourself is a significant, specialized operational burden — a managed service handles that for you, letting the team focus purely on deploying and operating application workloads.

Q18. What's the difference between SLA and an internal SLO?

A: An SLA is a formal, often contractual commitment made to customers/stakeholders; an SLO (Service Level Objective) is typically a stricter INTERNAL target a team holds itself to, giving buffer room before an SLA would actually be breached.

Q19. Why should alerts be based on user-facing symptoms rather than every internal metric?

A: Alerting on every minor internal fluctuation causes alert fatigue, where real, important alerts get ignored among noise — alerting specifically on symptoms that affect real users (error rate, latency, SLA breaches) keeps alerts meaningful and actionable.

Q20. How does an Ingress differ from a Kubernetes Service?

A: A Service provides internal (or basic external via LoadBalancer type) access to a single set of Pods; an Ingress sits in front of multiple Services, handling external HTTP(S) routing by hostname/path and TLS termination through one shared entry point.

Q21. Why is "forward-only" generally preferred over true database rollback?

A: Once real data has been written against a new schema, reversing that schema change can genuinely corrupt or lose that data — writing a new forward migration to correct course is usually safer than attempting to undo an already-applied change.

Q22. What real information does a good RCA document need to include beyond "what broke"?

A: The true underlying root cause (not just the immediate trigger), the full timeline of detection and response, and concrete, assigned action items that prevent the same CLASS of incident from recurring — not just a description of what happened.

Q23. Why is it important that developers understand basic cloud/DevOps concepts, even if they don't personally manage infrastructure?

A: It speeds up incident diagnosis (knowing where to look — logs, metrics, recent deployments), improves communication with DevOps/Cloud/Production Support teams, and leads to code/architecture decisions that are naturally more deployable and observable.

Q24. What's the difference between DEV, QA, UAT, and PROD in terms of who "owns" sign-off at each stage?

A: Developers own DEV; the QA/Test team owns formal testing sign-off in QA; business stakeholders/Product

Owners own acceptance sign-off in UAT; Release Management/Change Advisory Boards typically own the final go-live approval into PROD.

Q25. How would you investigate "the app is slow" reported with no other detail, using the tools covered in this guide?

A: Check API monitoring dashboards for the specific endpoint's latency trend, check distributed tracing (Application Insights) to see which layer/dependency is actually slow, and check recent deployments/Change Requests for anything correlating with when the slowness started.

Q26. Why do organizations separate the DBA team from application developers for schema changes?

A: DBAs bring specialized expertise in indexing strategy, migration safety at real production data scale, and backup/recovery planning — catching risks (like a table lock during a large migration, Section 84) that an application developer might not anticipate.

Q27. What's the real purpose of a staging/UAT environment if QA has already tested the feature?

A: UAT validates the feature against REAL business requirements and workflows from the people who actually requested it, catching genuine misunderstandings of intent that technical QA testing (correctness against a spec) might not surface.

Q28. Why might a hotfix pipeline skip certain stages that a normal release pipeline always runs?

A: To minimize time-to-resolution during an active incident — genuinely non-essential slow stages (full end-to-end suites) may be skipped, while critical safety checks (unit tests, security scans) are still enforced, balancing speed against not making things worse.

Q29. How does IAM's "least privilege" principle reduce real production risk?

A: Limiting each person/service to only the exact permissions they need shrinks the "blast radius" if an account is ever compromised or a mistake is made — a developer without production database write access simply CANNOT accidentally (or maliciously) corrupt production data directly.

Q30. Summarize the entire deployment pipeline covered in this guide in one sentence.

A: Code is committed via Git and reviewed through a Pull Request, built and quality-gated by a CI pipeline into a versioned container image stored in a registry, promoted through DEV/QA/UAT by a Release Pipeline with approval gates, run and auto-scaled by Kubernetes on cloud infrastructure, continuously watched by logging/monitoring tools, and supported in production by a structured incident/problem/change management process with clear team ownership at every layer.

88. Glossary of Terms

Term	Meaning
Artifact	A versioned build output (JAR, Docker image) passed between pipeline stages
Blue-Green Deployment	Running two full environments, switching all traffic at once for instant rollback capability
Canary Deployment	Releasing a new version to a small percentage of traffic first, before a full rollout
Control Plane	The set of components managing a Kubernetes cluster's overall state and scheduling decisions
Idempotency	A property where repeating an operation has the same effect as doing it once — critical for safe retries
Observability	The combination of logs, metrics, and traces that let you understand a running system's real behavior
Runbook	A documented, step-by-step guide for responding to a specific known type of incident/alert
Service Mesh	An infrastructure layer (e.g., Istio) managing service-to-service communication, security, and observability transparently
Throttling	Deliberately limiting request rate to protect a system from being overwhelmed
War Room	A dedicated, focused response session (physical or virtual) assembled during a major incident

89. Quick Revision Summary

- Every deployment topic is one STEP in a single connected pipeline: code → build → container → cluster → cloud → monitoring → support.
- "Build once, promote many" — the exact same tested artifact moves through DEV→QA→UAT→PROD, never rebuilt per environment.
- Kubernetes automates what Docker alone can't: scheduling, scaling, self-healing, and rolling updates across many machines.
- Rollback is almost always the fastest, safest first response to a production incident — deep root-cause investigation happens afterward via RCA.
- Secrets belong only in a dedicated secrets manager, never hardcoded or committed — and never in a frontend environment variable.
- Alerts should reflect real user-facing symptoms, not every internal metric blip, to avoid alert fatigue.
- Know exactly who owns what (Dev, DevOps, Cloud, Production Support, SRE, DBA, Network, Security, QA, Release Management) so you escalate correctly during an incident.
- A real production incident's true root cause usually differs from its immediate trigger — a good RCA finds and fixes the former, not just the latter.